# Digitized Chiral Soliton Lattice Hamiltonian

Generalization to arbitrary qubits per site and arbitrary lattice size

Extends the one-qubit, two-site derivation of *CSL — quantum computing* (eqs. 1.1–1.21) to $n_q$ qubits per site ($Q = 2^{n_q}$ field values) and $N$ sites, with explicit results for $(n_q, N) = (1,4)$, $(2,2)$ and $(2,4)$.

All formulae verified numerically — 57/57 checks, including entry-by-entry reproduction of eq. (1.21).

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## 1. Setup and the generalized Hamiltonian

The classical effective Hamiltonian density in one spatial dimension, with the magnetic shorthand $H \equiv \mu B/(4\pi^2 f_\pi^2)$, promotes to the quantum energy

$$\hat{E} = f_\pi^2 \int dx \left[\frac{\Pi^2}{2 f_\pi^4} + \tfrac{1}{2}(\partial_x \phi)^2 + m_\pi^2 (1 - \cos\phi) - H\,\partial_x \phi\right].$$

Discretizing on $N$ sites of spacing $a$, $x = na$ with $n = 1, \dots, N$, and compactifying the two $\phi$-derivative terms exactly as in eq. (1.6),

$$\tfrac{1}{2}(\Delta\phi_n)^2 \;\longrightarrow\; 1 - \cos\Delta\phi_n, \qquad \Delta\phi_n \;\longrightarrow\; \sin\Delta\phi_n, \qquad \Delta\phi_n \equiv \phi_{n+1} - \phi_n,$$

gives the general lattice Hamiltonian, valid for any number of sites and any field digitization:

$$\hat{H} = t \sum_{n=1}^{N} \hat{\Pi}_n^2 \;+\; J \sum_{\langle n,n+1\rangle} \left[1 - \cos\Delta\phi_n\right] \;+\; h \sum_{n=1}^{N} \left[1 - \cos\phi_n\right] \;-\; \kappa \sum_{\langle n,n+1\rangle} \sin\Delta\phi_n$$

$$t = \frac{1}{2 f_\pi^2 a}, \qquad J = \frac{f_\pi^2}{a}, \qquad h = f_\pi^2 m_\pi^2 a, \qquad \kappa = f_\pi^2 H = \frac{\mu B}{4\pi^2}.$$

The bond sum runs over $N-1$ bonds for open boundary conditions — the case used in eq. (1.20), where the single $N = 2$ bond appears once — or over $N$ bonds for periodic boundaries. Note that $\kappa$ carries no factor of $a$: the magnetic term is topological in origin, which is why it is the one coupling that does not scale with the lattice spacing.

## 2. Digitization: clock and shift operators

The compact angle $\phi \in [0, 2\pi)$ is replaced by $Q$ equally spaced points $\phi_k = 2\pi k/Q$, $k = 0, \dots, Q-1$, and the integer $k$ is stored in binary on the $n_q$-qubit site register, $k = \sum_{j=0}^{n_q-1} 2^j b_j$. Define the clock and shift operators

$$\hat{Z}_Q\,|k\rangle = \omega^k |k\rangle \;\; \left(= e^{i\hat{\phi}}\right), \qquad \hat{X}_Q\,|k\rangle = |k + 1 \bmod Q\rangle, \qquad \omega = e^{2\pi i/Q}.$$

Every term of the Hamiltonian is then *exact for any* $Q$:

| term | operator form |
| --- | --- |
| $\cos\phi_n$ | $\frac{1}{2}\left(\hat{Z}_n + \hat{Z}_n^\dagger\right) = \mathrm{Re}\,\hat{Z}_n$ |
| $\sin\phi_n$ | $\frac{1}{2i}\left(\hat{Z}_n - \hat{Z}_n^\dagger\right) = \mathrm{Im}\,\hat{Z}_n$ |
| $\cos\Delta\phi_n$ | $\frac{1}{2}\left(\hat{Z}_{n+1}\hat{Z}_n^\dagger + \hat{Z}_{n+1}^\dagger\hat{Z}_n\right)$ |
| $\sin\Delta\phi_n$ | $\frac{1}{2i}\left(\hat{Z}_{n+1}\hat{Z}_n^\dagger - \hat{Z}_{n+1}^\dagger\hat{Z}_n\right)$ |

For $Q = 2$ these collapse to $\cos\phi = Z$, $\cos\Delta\phi = Z_n Z_{n+1}$ and $\sin\Delta\phi = 0$, i.e. eqs. (1.10), (1.17) and (1.18) of the original note. The angle-addition step used there, $\cos(\phi_{n+1} - \phi_n) = \cos\phi_{n+1}\cos\phi_n + \sin\phi_{n+1}\sin\phi_n$, is simply the real part of $\hat{Z}_{n+1}\hat{Z}_n^\dagger$.

### 3. The kinetic term at arbitrary $Q$

This is the piece that requires genuine generalization: eq. (1.15) was obtained by a $2 \times 2$ similarity transformation, which does not extend. The conjugate momentum $\hat{\Pi} = -i\partial_\phi$ on the compact rotor has integer eigenvalues $l$, taken in the symmetric window

$$l \in \left\{-\tfrac{Q}{2}, \ldots, \tfrac{Q}{2} - 1\right\}, \qquad \text{which for } Q = 2 \text{ is } \{0, -1\} \text{ — eq. (1.11)},$$

with position and momentum bases related by the discrete Fourier transform $|l\rangle = Q^{-1/2} \sum_k \omega^{kl} |k\rangle$, eq. (1.12). Since $\hat{X}_Q |l\rangle = \omega^{-l} |l\rangle$, any function of $l$ is a polynomial in the shift operator. Writing $\hat{\Pi}^2 = \sum_l l^2 |l\rangle\langle l|$ in the position basis gives $\hat{\Pi}^2 = \sum_{m=0}^{Q-1} c_m \hat{X}_Q^m$ with $c_m = \frac{1}{Q} \sum_l l^2 \omega^{lm}$. Evaluating that sum with the generating function $f(x) = \sum_{l=-Q/2}^{Q/2-1} x^l$, which vanishes at $x = \omega^m$ for $m \not\equiv 0$,

$$c_m = \frac{1}{Q} \, (x\partial_x)^2 f \Big|_{x=\omega^m} = \frac{-2(-1)^m \, \omega^m}{(\omega^m - 1)^2} = \frac{(-1)^m}{2 \sin^2(\pi m/Q)},$$

while $\sum_l l^2 = \frac{1}{12} Q(Q-1)(Q-2) + \frac{1}{4} Q^2$ fixes the diagonal:

$$\boxed{\hat{\Pi}^2 = \frac{Q^2 + 2}{12} \mathbb{1} \; + \; \sum_{m=1}^{Q-1} \frac{(-1)^m}{2 \sin^2(\pi m/Q)} \, \hat{X}_Q^m.}$$

Hermiticity is automatic because $c_m = c_{Q-m}$ and $\hat{X}_Q^{Q-m} = (\hat{X}_Q^m)^\dagger$; the identity coefficient also satisfies $c_0 = -\sum_{m \neq 0} c_m$, which is just the statement that the $l = 0$ eigenvalue vanishes. Two independent checks:

- At $Q = 2$: $\hat{\Pi}^2 = \frac{1}{2}(1 - X)$, so $t\hat{\Pi}^2 = \frac{1}{4 f_\pi^2 a}(1 - X)$ — **exactly eq. (1.15)**.
- The diagonal element is $\langle k | \hat{\Pi}^2 | k \rangle = Q^2/12 + 1/6$, which is the Fornberg pseudo-spectral second-derivative matrix on a periodic grid — as it must be, since $\hat{\Pi}^2 = -\partial_\phi^2$.

At $Q = 4$ the expansion terminates after three terms, $\hat{\Pi}^2 = \frac{3}{2} \mathbb{1} - (\hat{X} + \hat{X}^\dagger) + \frac{1}{2} \hat{X}^2$, with spectrum $\{0, 1, 1, 4\} = l^2$ for $l \in \{0, \pm 1, -2\}$.

### 4. Qubit encoding

Under the binary encoding the clock operator factorizes into single-qubit phase gates, and the top qubit contributes a bare $Z$:

$$\hat{Z}_Q = \bigotimes_{j=0}^{n_q - 1} \begin{pmatrix} 1 & 0 \\ 0 & \omega^{2^j} \end{pmatrix}_j = i \, e^{-i\pi/Q} \, Z_{n_q - 1} \prod_{j=0}^{n_q - 2} \left[ \cos \tfrac{\pi 2^j}{Q} \, \mathbb{1} - i \sin \tfrac{\pi 2^j}{Q} \, Z_j \right].$$

Expanding the product over subsets $S \subseteq \{0, \ldots, n_q - 2\}$ gives closed-form Pauli series for both trigonometric functions,

$$\cos \hat{\phi} = \sum_S \sin\!\Big(\tfrac{\pi}{Q} + |S| \tfrac{\pi}{2}\Big) A_S \; Z_{n_q - 1} \prod_{j \in S} Z_j, \qquad \sin \hat{\phi} = \sum_S \cos\!\Big(\tfrac{\pi}{Q} + |S| \tfrac{\pi}{2}\Big) A_S \; Z_{n_q - 1} \prod_{j \in S} Z_j,$$

$$A_S = \prod_{j \in S} \sin \tfrac{\pi 2^j}{Q} \; \prod_{j \notin S} \cos \tfrac{\pi 2^j}{Q}.$$

Each carries exactly $Q/2$ Pauli-$Z$ strings, every one of them containing $Z_{n_q - 1}$ — the shift $\phi \to \phi + \pi$ flips the most significant bit and flips the sign of both functions, so neither can contain a string that is even under it.

The shift $\hat{X}_Q$ is the mod-$Q$ increment, a cascade of multi-controlled NOTs. Its Pauli expansion has $3^{n_q - 1} + 1$ strings, so for $n_q \gtrsim 3$ one should not expand it at all: implement $e^{-i\tau t \hat{\Pi}^2}$ as $\mathrm{QFT} \to$ diagonal $e^{-i\tau t \, l^2}$ phases $\to$

QFT$^\dagger$, costing $O(n_q^2)$ gates.

***The one-qubit case,*** $n_q = 1$ ***(***$Q = 2$***)***

$$\cos\phi_n = Z_n, \qquad \sin\phi_n = 0, \qquad \hat{\Pi}_n^2 = \tfrac{1}{2}(1 - X_n).$$

***The two-qubit case,*** $n_q = 2$ ***(***$Q = 4$***)***

Writing $Z_{n,1}$ and $Z_{n,0}$ for the most and least significant qubit of site $n$ (so $\phi = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ for $|b_1 b_0\rangle = |00\rangle, |01\rangle, |10\rangle, |11\rangle$):

$$\cos\phi_n = \tfrac{1}{2}\big(Z_{n,1} + Z_{n,1}Z_{n,0}\big), \qquad \sin\phi_n = \tfrac{1}{2}\big(Z_{n,1} - Z_{n,1}Z_{n,0}\big),$$
$$\hat{\Pi}_n^2 = \tfrac{3}{2}1 - X_{n,0} - X_{n,1}X_{n,0} + \tfrac{1}{2}X_{n,1},$$
$$\cos\Delta\phi_n = \tfrac{1}{2}\big(Z_{n,1}Z_{n+1,1} + Z_{n,1}Z_{n,0}Z_{n+1,1}Z_{n+1,0}\big),$$
$$\sin\Delta\phi_n = \tfrac{1}{2}\,Z_{n,1}Z_{n+1,1}\big(Z_{n,0} - Z_{n+1,0}\big).$$

The mass term reads $m_\pi^2 a\,\mathrm{diag}(0,1,2,1)$, the $Q = 4$ analogue of eq. (1.9), and $\hat{\Pi}_n^2$ above has the required spectrum $\{0, 1, 1, 4\}$.

> **Physics remark.** At $Q = 2$ one has $\Delta\phi \in \{0, \pm\pi\}$, so $\sin\Delta\phi \equiv 0$ and the coupling $\kappa$ *drops out of the Hamiltonian identically, at any* $N$. Eq. (1.18) is therefore not an artifact of the two-site truncation: a one-qubit register cannot see the magnetic field at all, and no chiral soliton lattice can form. The CSL first becomes accessible at $n_q = 2$. Relatedly, the linear term $-\kappa\sum_n \Delta\phi_n$ telescopes to a boundary term, whereas the compactified $\sin\Delta\phi_n$ does not — and it is exactly that failure to telescope which allows the soliton lattice to form.

## 5. Results

Sites are labelled $n = 1, \dots, N$ with open boundary conditions. All coefficients are exact rationals times $t$, $J$, $h$, $\kappa$.

***Reference:*** $n_q = 1$***,*** $N = 2$ ***— the original case***

$$\hat{H} = t\sum_{n=1}^{2} \tfrac{1}{2}\big(1 - X_n\big) + J\big(1 - Z_1 Z_2\big) + h\sum_{n=1}^{2}\big(1 - Z_n\big)$$

With $t = 1/(2f_\pi^2 a)$ this reproduces eq. (1.20), and its matrix reproduces eq. (1.21) entry by entry.

### A. *One qubit, four sites:* $n_q = 1$, $N = 4$

4 qubits, $16 \times 16$, 12 Pauli strings, $\kappa$ absent.

$$\boxed{\hat{H} = (2t + 3J + 4h)\,1 - \frac{t}{2}\sum_{n=1}^{4} X_n - J\sum_{n=1}^{3} Z_n Z_{n+1} - h\sum_{n=1}^{4} Z_n}$$

This is an open transverse-field Ising chain with a longitudinal field. In the computational basis the diagonal element of the configuration $(k_1 \dots k_4)$, $k_n \in \{0,1\}$, is $2t + 2J\,\#\{n : k_n \neq k_{n+1}\} + 2h\,\#\{n : k_n = 1\}$, and the only off-diagonal elements are $-t/2$ between configurations differing at a single site.

### B. *Two qubits, two sites:* $n_q = 2$, $N = 2$

4 qubits, $16 \times 16$, 15 Pauli strings. This is the smallest system in which the magnetic term is non-trivial.

$$\hat{H} = (3t + J + 2h)\,1 + t\sum_{n=1}^{2}\Big[-X_{n,0} + \tfrac{1}{2}X_{n,1} - X_{n,1}X_{n,0}\Big]$$
$$-\frac{J}{2}\Big[Z_{1,1}Z_{2,1} + Z_{1,1}Z_{1,0}Z_{2,1}Z_{2,0}\Big] - \frac{h}{2}\sum_{n=1}^{2}\Big[Z_{n,1} + Z_{n,1}Z_{n,0}\Big]$$
$$+\frac{\kappa}{2}\Big[Z_{1,1}Z_{2,1}Z_{2,0} - Z_{1,1}Z_{1,0}Z_{2,1}\Big]$$

Equivalently, in the basis $|k_1 k_2\rangle$ the diagonal is

$$3t + J\Big[1 - \cos\tfrac{\pi(k_2-k_1)}{2}\Big] + h\sum_{n=1}^{2}\Big[1 - \cos\tfrac{\pi k_n}{2}\Big] - \kappa\sin\tfrac{\pi(k_2-k_1)}{2},$$

with off-diagonal elements $-t$ for $\Delta k = \pm 1$ and $+t/2$ for $\Delta k = \pm 2$ on a single site.

### *C. Two qubits, four sites:* $n_q = 2$, $N = 4$

8 qubits, $256 \times 256$, 33 Pauli strings, maximum Pauli weight 4.

$$\hat{H} = (6t + 3J + 4h)\,1 + t\sum_{n=1}^{4}\Big[-X_{n,0} + \tfrac{1}{2}X_{n,1} - X_{n,1}X_{n,0}\Big]$$
$$-\frac{J}{2}\sum_{n=1}^{3}\Big[Z_{n,1}Z_{n+1,1} + Z_{n,1}Z_{n,0}Z_{n+1,1}Z_{n+1,0}\Big] - \frac{h}{2}\sum_{n=1}^{4}\Big[Z_{n,1} + Z_{n,1}Z_{n,0}\Big]$$
$$-\frac{\kappa}{2}\sum_{n=1}^{3}Z_{n,1}Z_{n+1,1}\Big[Z_{n,0} - Z_{n+1,0}\Big]$$

The 32 non-identity strings split as 12 kinetic (one- and two-local $X$), 6 gradient (two- and four-local $Z$), 8 mass (one- and two-local $Z$) and 6 magnetic (three-local $Z$). The whole kinetic sector mutually commutes and the whole potential sector is diagonal, so a Trotter step needs only two layers.

## 6. Resource scaling

Non-identity Pauli strings for general $n_q$ and $N$, open boundary conditions:

| sector | strings | max weight | $n_q$=1 | $n_q$=2 | $n_q$=3 |
|---|---|---|---|---|---|
| kinetic ($X$) | $N \cdot 3^{n_q-1}$ | $n_q$ | $N$ | $3N$ | $9N$ |
| mass ($Z$) | $N \cdot Q/2$ | $n_q$ | $N$ | $2N$ | $4N$ |
| gradient ($Z$) | $(N-1)\,2^{2n_q-3}$ | $2n_q$ | $N-1$ | $2(N{-}1)$ | $8(N{-}1)$ |
| magnetic ($Z$) | $(N-1)\,2^{2n_q-3}$ | $2n_q-1$ | 0 | $2(N{-}1)$ | $8(N{-}1)$ |

The bond counts follow because $\cos\phi$ and $\sin\phi$ each carry $Q/2$ strings, so $\cos\Delta\phi_n = C_{n+1}C_n + S_{n+1}S_n$ starts from $2(Q/2)^2$ products of which exactly half cancel. The $n_q = 1$ column is the degenerate case: $\sin\phi \equiv 0$ collapses the gradient term to one string per bond and kills the magnetic term entirely. Total qubits $Nn_q$; Hilbert dimension $2^{Nn_q} = Q^N$.

## 7. Numerics

Exact diagonalization with $f_\pi = a = m_\pi = 1$, so $t = \frac{1}{2}$ and $J = h = 1$, open boundaries. Listed are the ground-state energy $E_0$ and the CSL order parameter $\langle s \rangle \equiv \langle \sin \Delta\phi \rangle$ averaged over bonds.

| $\kappa$ | $E_0\ (1,4)$ | $\langle s \rangle$ | $E_0\ (2,2)$ | $\langle s \rangle$ | $E_0\ (2,4)$ | $\langle s \rangle$ |
|---|---|---|---|---|---|---|
| 0.0 | 0.9480 | 0 | 1.0807 | 0.0000 | 2.2809 | 0.0000 |
| 1.0 | 0.9480 | 0 | 1.0016 | 0.1760 | 2.1834 | 0.0776 |
| 2.0 | 0.9480 | 0 | 0.6666 | 0.5192 | 1.6014 | 0.3616 |
| 3.0 | 0.9480 | 0 | −0.0104 | 0.7967 | 0.0303 | 0.6667 |

Columns are labelled by $(n_q, N)$. The one-qubit column is exactly flat in $\kappa$, as it must be; at $n_q = 2$ the order parameter turns on and the spectral gap closes as the field is increased.

## 8. Verification

The accompanying code runs 57 independent checks, all passing:

- The $n_q = 1$, $N = 2$ Hamiltonian matrix matches eq. (1.21) entry by entry.
- The $Q = 2$ limits of every general formula reproduce eqs. (1.10), (1.15), (1.17), (1.18).
- $\hat{\Pi}^2 = \sum_m c_m \hat{X}^m$ has spectrum exactly $\{l^2\}$ for $Q$ up to 32, and diagonal $Q^2/12 + 1/6$.
- Both product forms of $\hat{Z}_Q$, and the closed-form Pauli series for $\cos\phi$ and $\sin\phi$, verified up to $n_q = 5$.
- The $Q = 4$ operator identities of §4, and Hermiticity plus reality of all Pauli coefficients for the three cases of §5.

`csl_lattice.py` — general construction for any $(n_q, N)$, boundary condition and couplings. `verify.py` — the 57 checks. `cases.py` — Pauli-string Hamiltonians. `matrices.py` — explicit symbolic matrices. `spectra.py` — spectra and order parameter.